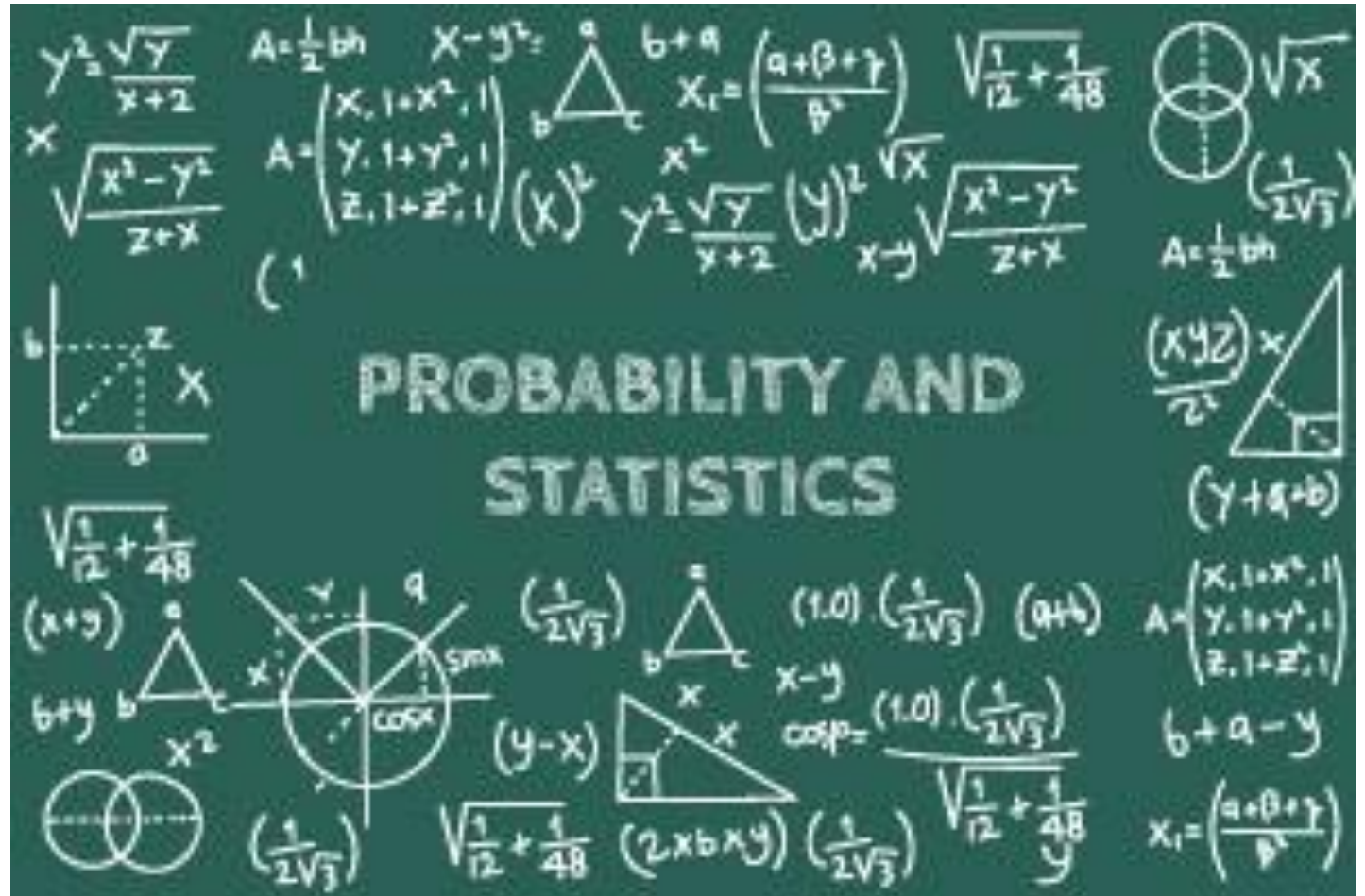


Lec # 03
ES111

Counting Sample Points

Book Reading: Sec. 2.3

@M. Tahir Naseem



Discussion Plan

1. Summary

2. **Counting** Sample points

3. **Practice** Problems (Sec. 2.3)



To calculate probability

To calculate probability

1. calculate "sample space"

To calculate probability

1. calculate "sample space"
2. Identify "event" of interest

To calculate probability

1. calculate "sample space"
2. Identify "event" of interest
3. Divide "event" by "sample space" $\longrightarrow$ probability

To calculate probability

1. calculate "sample space"  Today's class
2. Identify "event" of interest
3. Divide "event" by "sample space"  probability

Sample Space

Sample Space



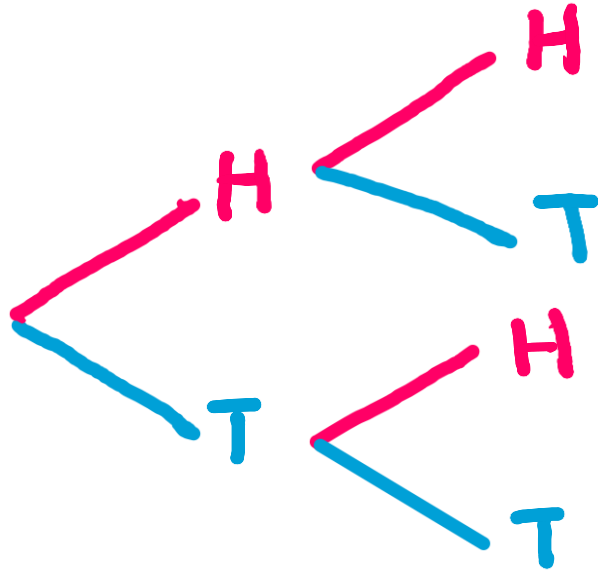
A set of all possible outcomes of statistical experiment is called sample space



Consider the experiment of tossing a **coin two times**.
Write the **sample space**.



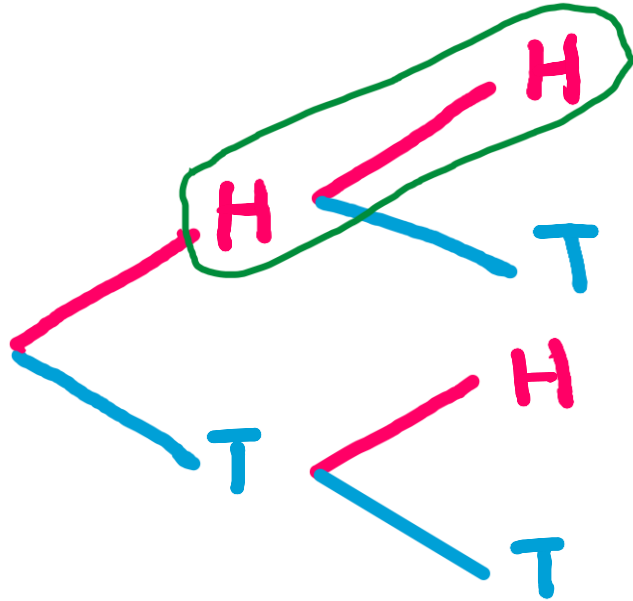
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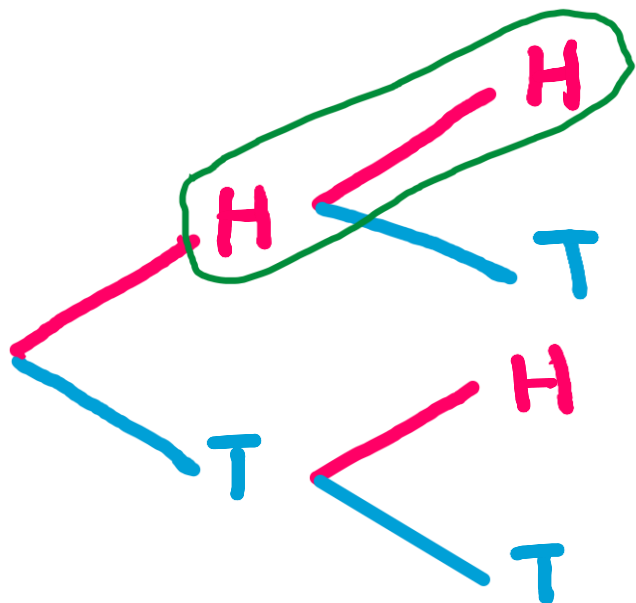
$$S = \{ \quad \}$$





Consider the experiment of tossing a **coin two times**.
Write the **sample space**.

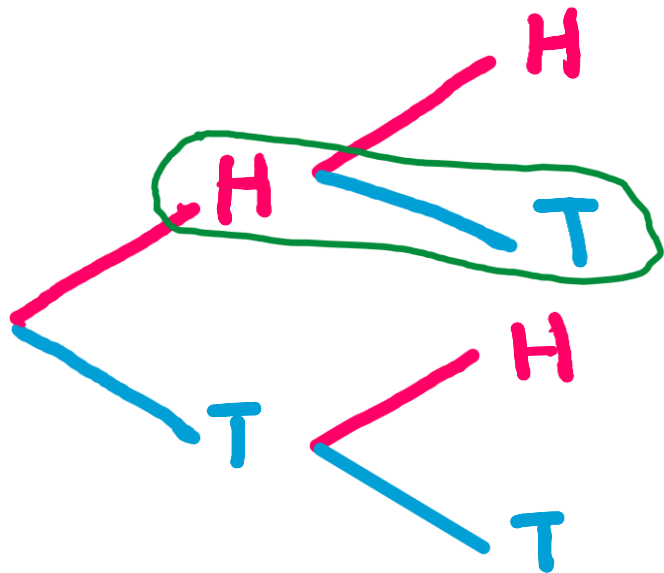
$$S = \{HH, HT, TH, TT\}$$





Consider the experiment of tossing a **coin two times**.
Write the **sample space**.

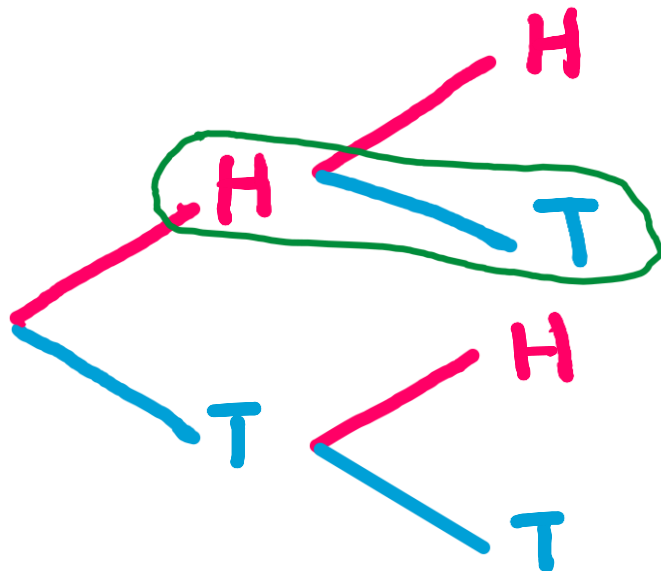
$$S = \{HH, HT, TH, TT\}$$





Consider the experiment of tossing a **coin two times**.
Write the **sample space**.

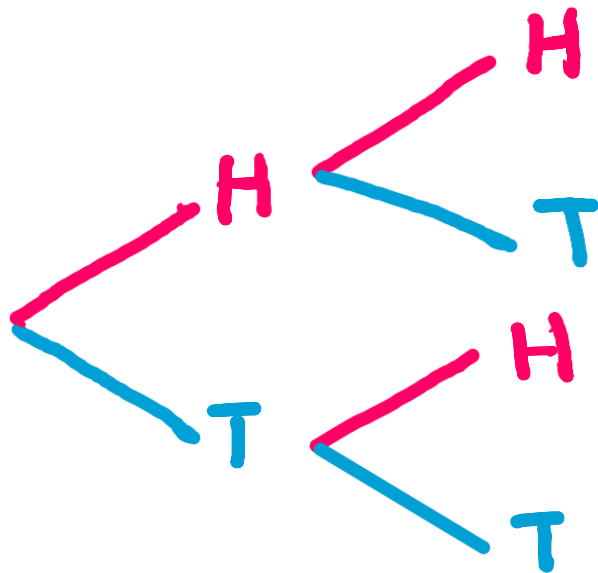
$$S = \{HH, HT, \quad \}$$





Consider the experiment of tossing a **coin two times**.
Write the **sample space**.

$$S = \{HH, HT, TH, TT\}$$



Events

Events

An event is a subset of sample space.

Counting Sample points

Counting Sample points

"How many different ways we can order?"



Counting Sample points

"How many different ways we can order?"



2

x

1

Counting Sample points

"How many different ways we can order?"



X

1

Counting Sample points

"How many different ways we can order?"



3

X

1

Counting Sample points

"How many different ways we can order?"



3

X

Counting Sample points

"How many different ways we can order?"



3

X



2

Counting Sample points

"How many different ways we can order?"



3

X



2

X



Counting Sample points

"How many different ways we can order?"



3

X



2

X



4

Counting Sample points

Multiplication Rule



3

x



2

x



4

Counting Sample points

Multiplication Rule

categories



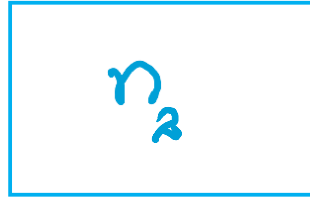
...



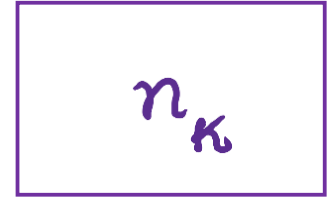
Counting Sample points

Multiplication Rule

Categories



...



All possibilities = $n_1 \times n_2 \cdots n_k$



How many sample points are there in the sample space when a pair of dice is thrown once?



How many sample points are there in the sample space when a **pair of dice** is thrown once?





How many sample points are there in the sample space when a **pair of dice** is thrown once?



$$n_1 = 6$$



$$n_2 = 6$$



How many sample points are there in the sample space when a **pair of dice** is thrown once?



$$n_1 = 6$$



$$n_2 = 6$$

$$= n_1 \times n_2 = 6 \times 6$$



Sam is going to assemble a computer by himself. He has the choice of **chips from two brands**, a **hard drive from four**, **memory from three**, and **an accessory bundle from five local stores**. How many different ways can Sam order the parts?



Sam is going to assemble a computer by himself. He has the choice of **chips from two brands**, a **hard drive from four**, **memory from three**, and **an accessory bundle from five local stores**. How many different ways can Sam order the parts?

$$= 2 \times 4 \times 3 \times 5 = 120$$

Ans



A certain brand of shoes comes in 5 different styles, with each style available in 4 distinct colors. If the store wishes to display pairs of these shoes showing all of its various styles and colors, how many different pairs will the store have on display?



A certain brand of shoes comes in **5 different styles**, with each style available **in 4 distinct colors**. If the store wishes to display pairs of these shoes showing all of its various styles and colors, how many different pairs will the store have on display?

$$= 5 \times 4 = 20$$

Ans

Counting Sample points

Multiplication Rule
↓

Repetition of Events



In how many different ways can a true-false test consisting of 9 questions be answered?



In how many different ways can a **true-false** test consisting of **9 questions** be answered?



In how many different ways can a **true-false** test consisting of **9 questions** be answered?

$$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$$



In how many different ways can a **true-false** test consisting of **9 questions** be answered?

$$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^9$$



If a multiple-choice test consists of 5 questions, each with 4 possible answers of which only 1 is correct, in how many different ways can a student check off one answer to each question?



If a multiple-choice test consists of **5 questions**, each **with 4 possible answers** of which only 1 is correct, in how many different ways can a student check off one answer to each question?

$$= 4^5$$

Ans

Counting Sample points



1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

Combinations

Combinations → Arrangements in which order does not matter.

Combinations → Arrangements in which order does not matter.
Arrangement in which order matters → permutations

Combinations

→ Arrangements in which order does not matter.

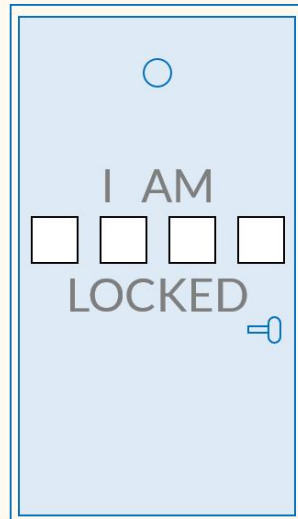
Arrangement in which order matters → permutations

Confused

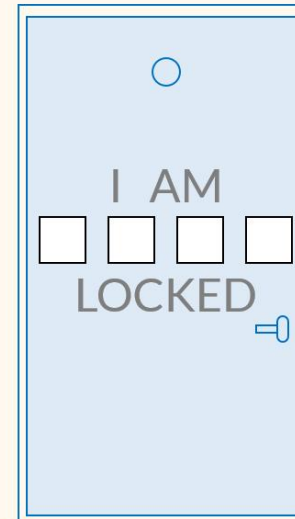


Should I unlock with Permutation or Combination ?

Permutation



Combination



Combinations → Arrangements in which order does not matter.

A

B

C

same

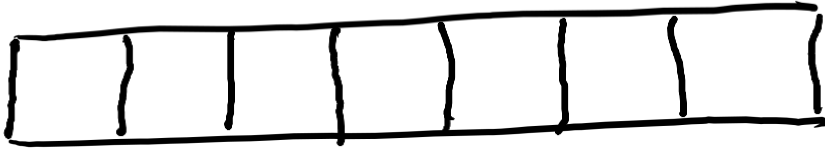
B

A

C

Combinations

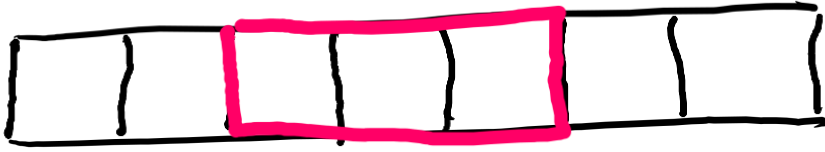
n outcomes



Combinations

γ outcomes

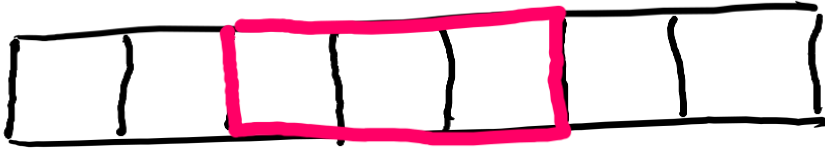
n outcomes



Combinations

r outcomes

n outcomes

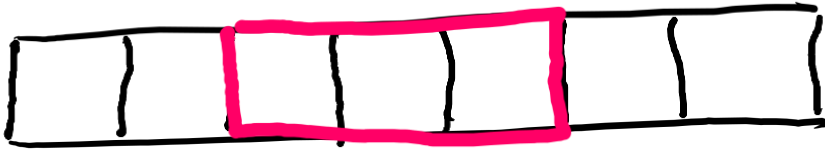


$$\text{combination} = \frac{n!}{r! (n-r)!}$$

Combinations

r outcomes

n outcomes

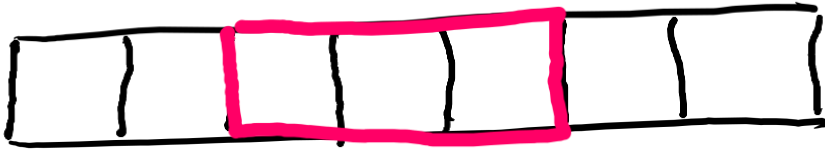


$$\text{combination} = \frac{n!}{r! (n-r)!} = {}^n C_r$$

Combinations

r outcomes

n outcomes



$$\text{combination} = \frac{n!}{r! (n-r)!} = {}^nC_r = \binom{n}{r}$$

Counting Sample points



1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

2.1 "no constraints"

Counting Sample points



1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$



In one year, three same awards will be given to a class of 25 graduate students in a statistics department. If each student can receive at most one award, how many possible selections are there?



In one year, **three same awards** will be given to a class of **25 graduate** students in a statistics department. If each student can receive at most one award, **how many possible** selections are there?



In one year, **three same awards** will be given to a class of **25 graduate** students in a statistics department. If each student can receive at most one award, **how many possible** selections are there?

$${}_nC_r = \frac{n!}{r! (n-r)!}$$



In one year, **three same awards** will be given to a class of **25 graduate** students in a statistics department. If each student can receive at most one award, **how many possible** selections are there?

$$n = 25$$
$$r = 3$$

$${}_nC_r = \frac{n!}{r! (n-r)!} = \frac{25!}{3! (25-3)!} \quad \underline{\text{Ans}}$$



How many ways are there to select 3 candidates from 8 equally qualified recent graduates for openings in an accounting firm?



How many ways are there to **select 3 candidates** from **8 equally qualified** recent graduates for openings in an accounting firm?



How many ways are there to **select 3 candidates** from **8 equally qualified** recent graduates for openings in an accounting firm?

$${}_8C_3 = \frac{8!}{3!(8-3)!}$$

$${}_nC_r = \frac{n!}{r!(n-r)!}$$



How many ways are there to **select 3 candidates** from **8 equally qualified** recent graduates for openings in an accounting firm?

$${}_nC_r = \frac{n!}{r! (n-r)!}$$

$${}_8C_3 = \frac{8!}{3! (8-3)!} = \frac{8!}{3! 5!}$$



How many ways are there to **select 3 candidates** from **8 equally qualified** recent graduates for openings in an accounting firm?

$${}_nC_r = \frac{n!}{r! (n-r)!}$$

$$\begin{aligned} {}_8C_3 &= \frac{8!}{3! (8-3)!} = \frac{8!}{3! 5!} \\ &= \frac{8 \times 7 \times 6 \times 5!}{3! 5!} \end{aligned}$$



How many ways are there to **select 3 candidates** from **8 equally qualified** recent graduates for openings in an accounting firm?

$${}_nC_r = \frac{n!}{r! (n-r)!}$$

$$\begin{aligned} {}_8C_3 &= \frac{8!}{3! (8-3)!} = \frac{8!}{3! 5!} \\ &= \frac{8 \times 7 \times \cancel{6} \times \cancel{5}!}{\cancel{3!} \cancel{5!}} \end{aligned}$$



How many ways are there to **select 3 candidates** from **8 equally qualified** recent graduates for openings in an accounting firm?

$${}_nC_r = \frac{n!}{r! (n-r)!}$$

$$\begin{aligned} {}_8C_3 &= \frac{8!}{3! (8-3)!} = \frac{8!}{3! 5!} \\ &= \frac{8 \times 7 \times \cancel{6} \times \cancel{5}!}{\cancel{3!} \cancel{5!}} = 56 \text{ ans} \end{aligned}$$

Counting Sample points

✓
1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

✓
2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

✓ 2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

2.2 no constraints but "selection is made from different categories"

Combinations

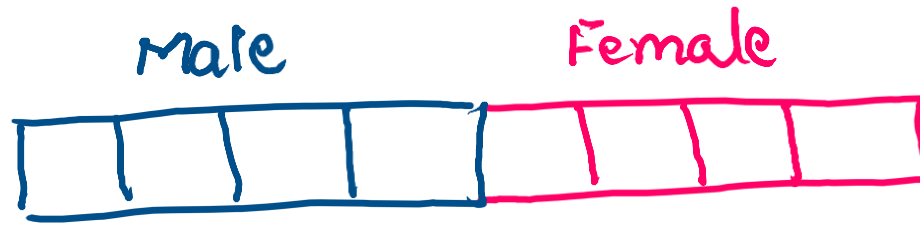
2.2

no constraints but "selection is made from different categories"

Combinations

2.2

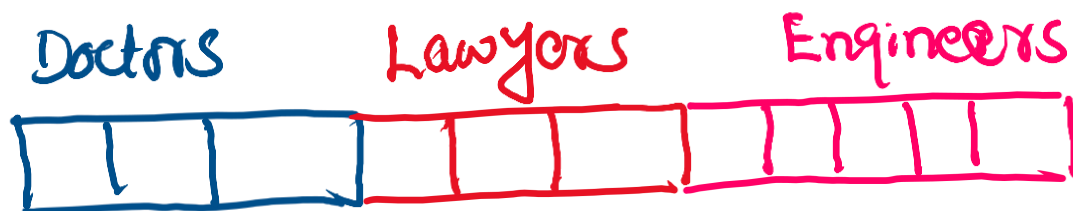
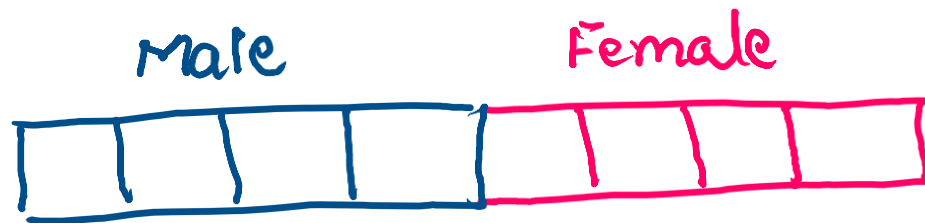
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Combinations

2.2

no constraints but "selection is made from different categories"

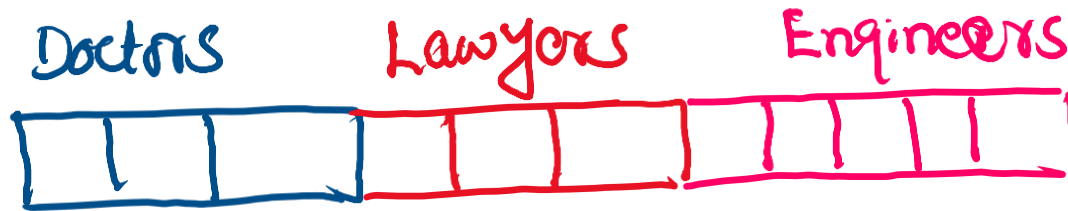
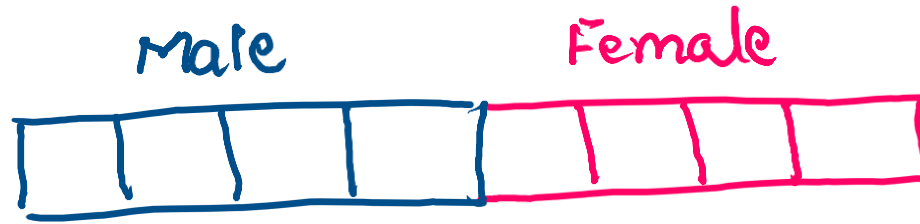


Combinations

2.2

no constraints but "selection is made from different categories"

→ "select a team of 3"

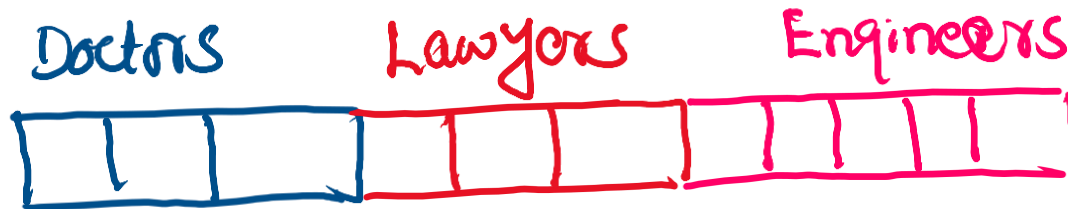
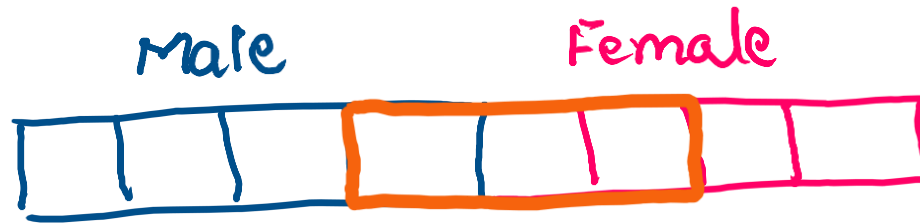


Combinations

2.2

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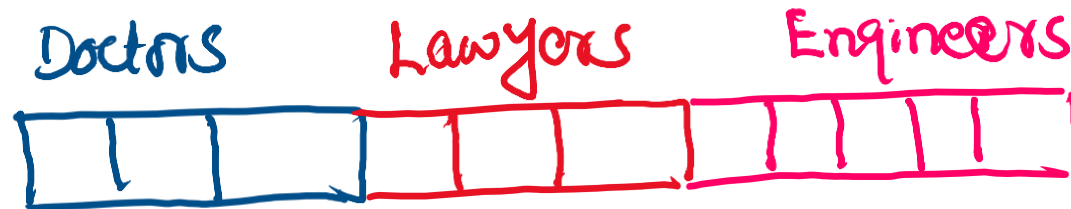
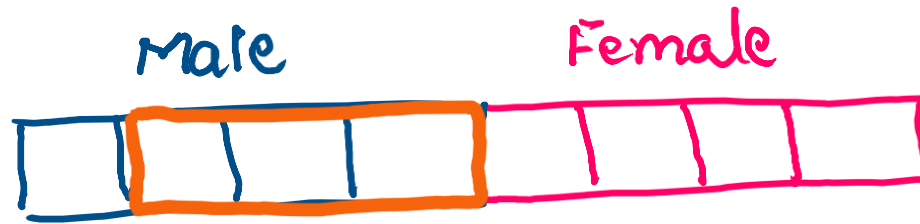


Combinations

2.2

no constraints but "selection is made from different categories"

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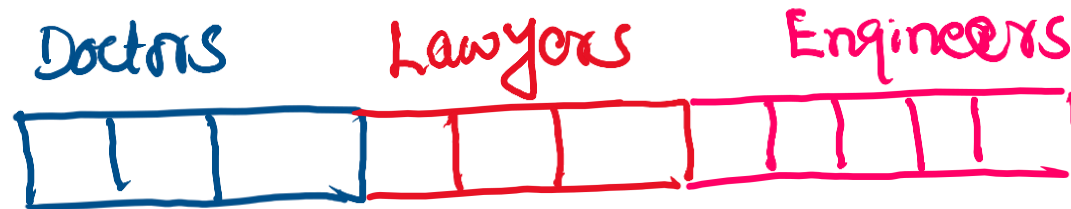
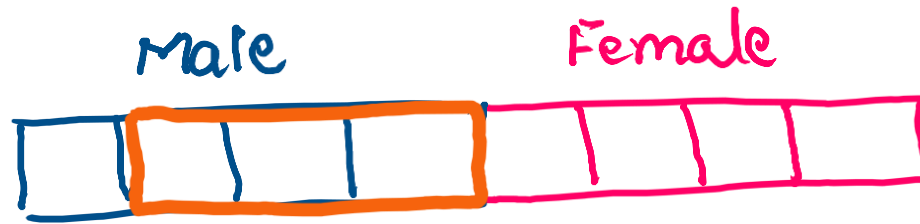


Combinations

2.2

no constraints but "selection is made from different categories"

→ "select a team of 3"



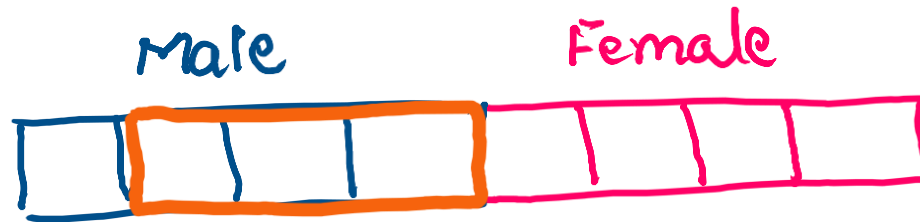
"no constraint over selection"

Combinations

2.2

no constraints but "selection is made from different categories"

→ "select a team of 3"



"no constraint over selection"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$



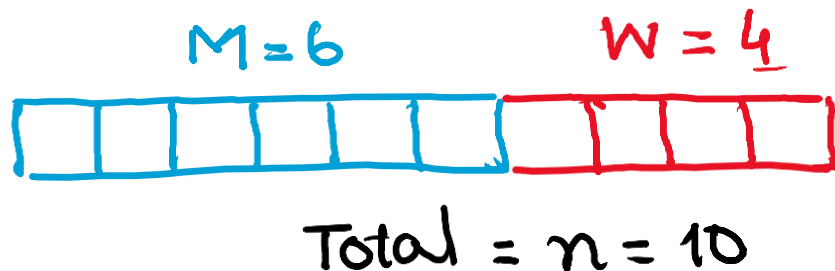
A committee of 5 people is to be formed from a group of 6 men and 4 women. How many ways it is possible to form a committee.



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee.



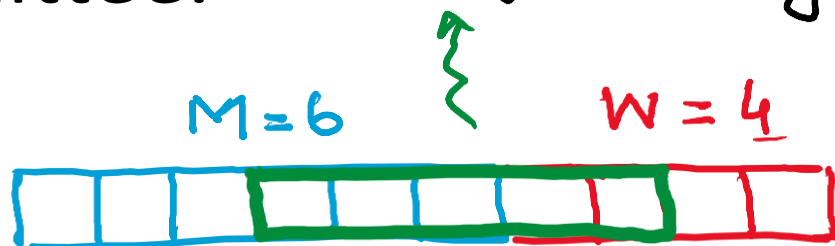
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A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee.

'can be any 5 shown as example'

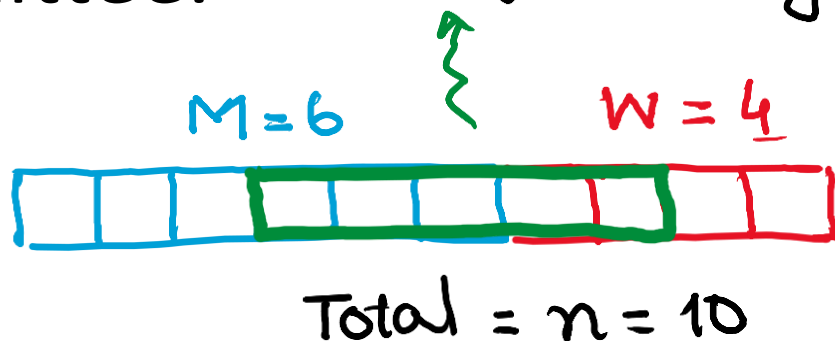


Total = $n = 10$



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee.

'can be any 5 shown as example'

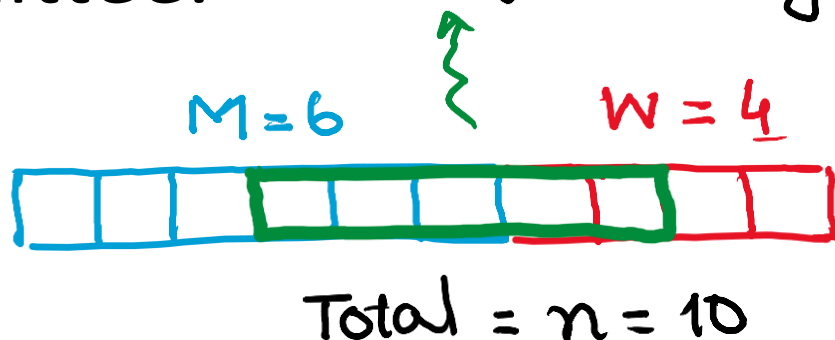


$${}_n C_r = \frac{n!}{r! (n-r)!}$$



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee.

'can be any 5 shown as example'



$${}_nC_r = \frac{n!}{r! (n-r)!}$$

$${}_{10}C_5 = \frac{10!}{5! (10-5)!}$$

Ans



If 3 books are picked at random from a shelf containing 5 novels, 3 books of poems, and a dictionary. In how many ways books can be picked.



If **3 books** are picked at random from a shelf containing **5 novels**, **3 books of poems**, and **a dictionary**. In how many ways books can be picked.



If **3 books** are picked at random from a shelf containing **5 novels**, **3 books of poems**, and **a dictionary**. In how many ways books can be picked.

$${}^nC_r = \frac{n!}{r! (n-r)!}$$



If **3 books** are picked at random from a shelf containing **5 novels**, **3 books of poems**, and **a dictionary**. In how many ways books can be picked.

$${}^9C_3 = \frac{9!}{3! (9-3)!}$$

Ans

$${}_nC_r = \frac{n!}{r! (n-r)!}$$

$$n = 9$$

$$r = 3$$

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

✓ 2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

Same

✓ 2.2 no constraints but "selection is made from different categories"

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

✓ 2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

Same

✓ 2.2 no constraints but "selection is made from different categories"

2.3 with constraints

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

✓ 2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

Same

✓ 2.2 no constraints but "selection is made from different categories"

"constraint selection from each different group"

2.3 with constraints

Combinations

2.3

with constraints

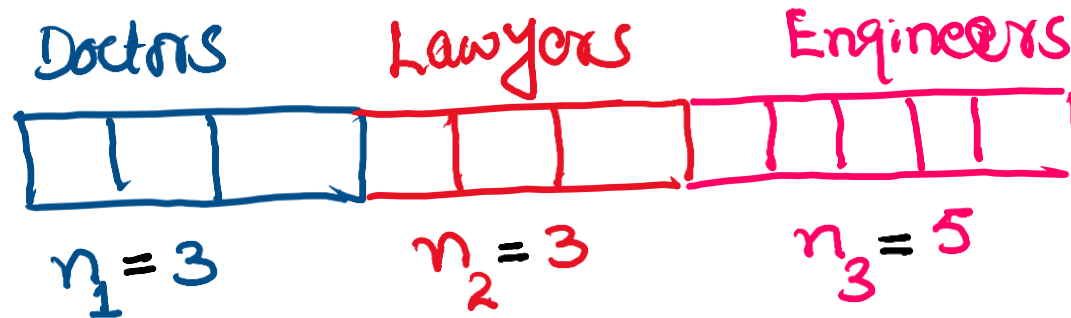
"constraint selection from each different group"

Combinations

(2.3)

with constraints

"constraint selection from each different group"

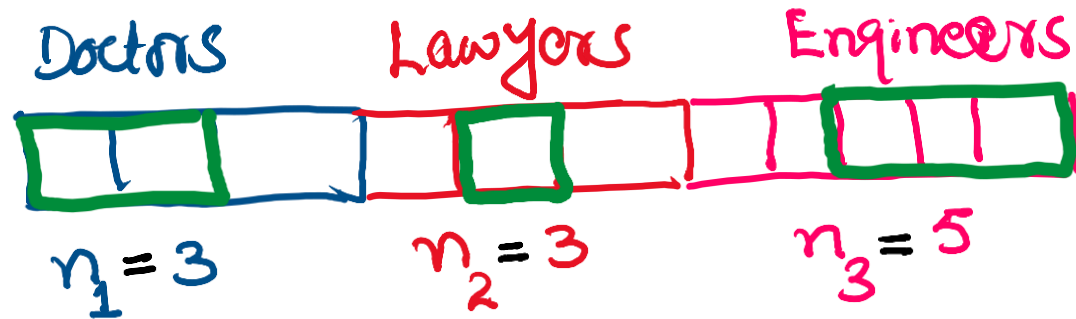


Combinations

(2.3) with constraints

"constraint selection from each different group"

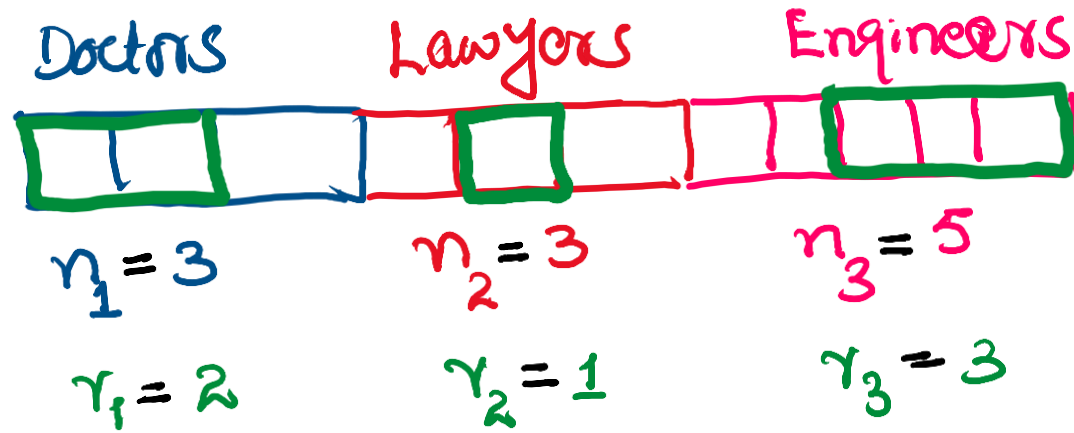
suppose Q "select 2 doctors 1 lawyer and 3 engineers"



Combinations

(2.3) with constraints $\rightarrow$ "constraint selection from each different group"

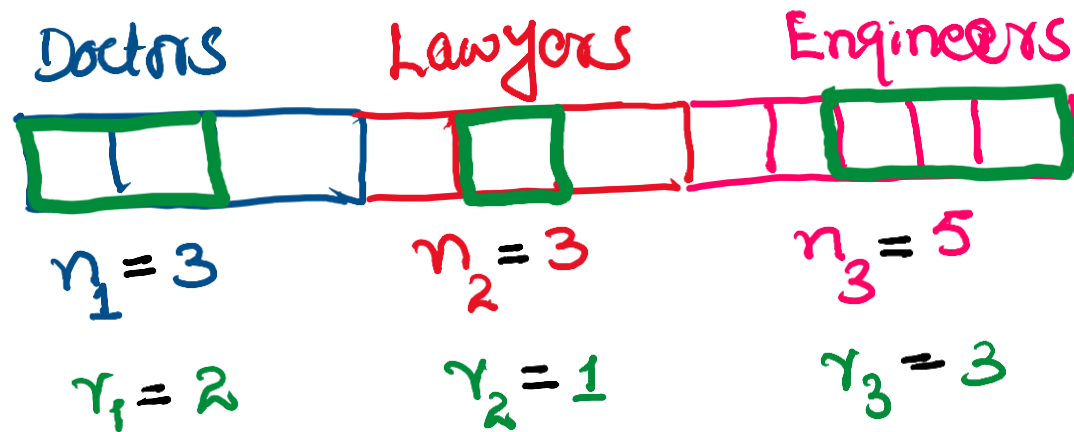
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Combinations

(2.3) with constraints $\rightarrow$ "constraint selection from each different group"

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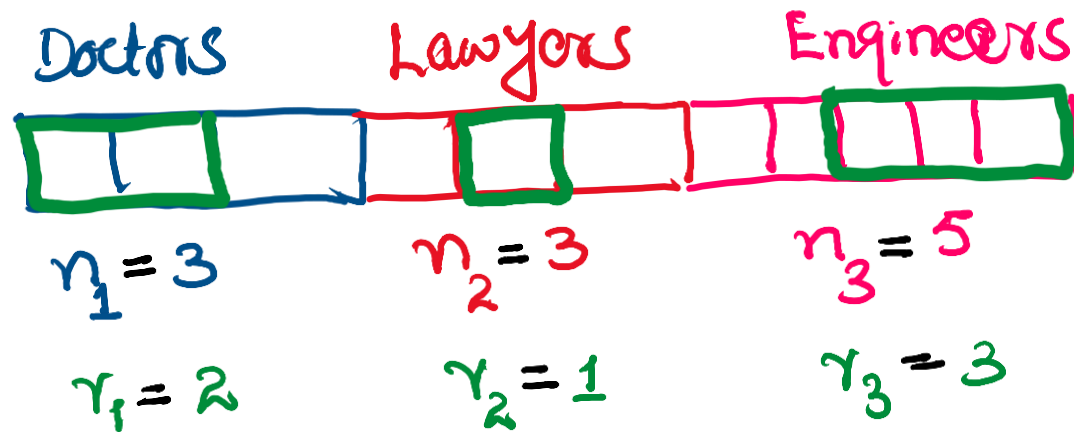


$${}_{n_1}C_{r_1}$$

Combinations

(2.3) with constraints $\rightarrow$ "constraint selection from each different group"

suppose Q "select 2 doctors 1 lawyer and 3 engineers"



$${}_{n_1}C_{r_1}$$

$${}_{n_2}C_{r_2}$$

Combinations

(2.3) with constraints

"constraint selection from each different group"

suppose Q "select 2 doctors 1 lawyer and 3 engineers"



$$n_1 = 3$$

$$r_1 = 2$$

$${}_{n_1}C_{r_1}$$

$$n_2 = 3$$

$$r_2 = 1$$

$${}_{n_2}C_{r_2}$$

$$n_3 = 5$$

$$r_3 = 3$$

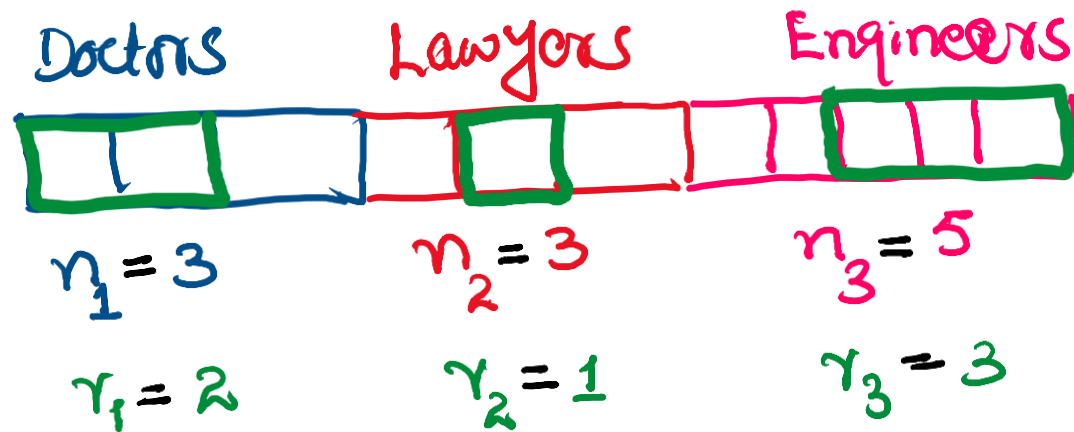
$${}_{n_3}C_{r_3}$$

Combinations

(2.3) with constraints

"constraint selection from each different group"

suppose Q "select 2 doctors 1 lawyer and 3 engineers"



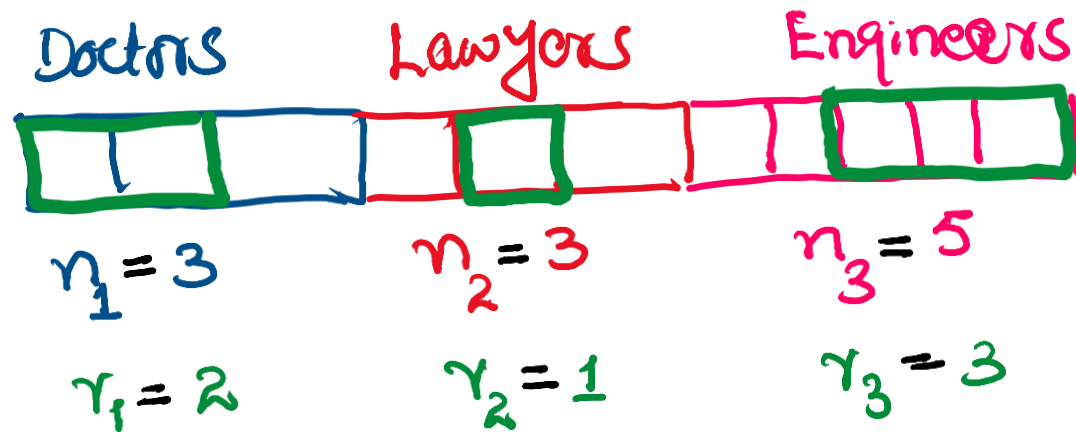
$$\text{Total possibilities} = {}^{n_1}C_{r_1} \times {}^{n_2}C_{r_2} \times {}^{n_3}C_{r_3}$$

Combinations

(2.3) with constraints

"constraint selection from each different group"

suppose Q "select 2 doctors 1 lawyer and 3 engineers"



$$\begin{aligned} \text{Total possibilities} &= {}^{n_1}C_{r_1} \times {}^{n_2}C_{r_2} \times {}^{n_3}C_{r_3} \\ &= {}^3C_2 \times {}^3C_1 \times {}^5C_3 \end{aligned}$$



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee.

we have solved earlier
notice no constraint!

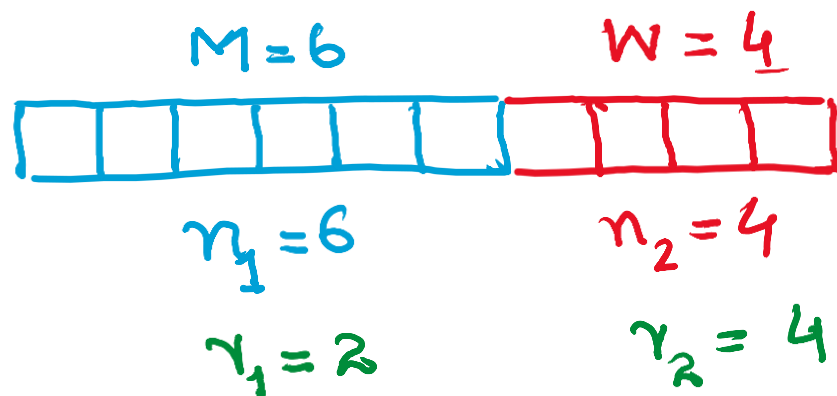


A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee if there are 2 men and 4 women.

→ "constraint added"



A committee of **5 people** is to be formed from a group of **6 men** and **4 women**. How many ways it is possible to form a committee if there are 2 men and 4 women.

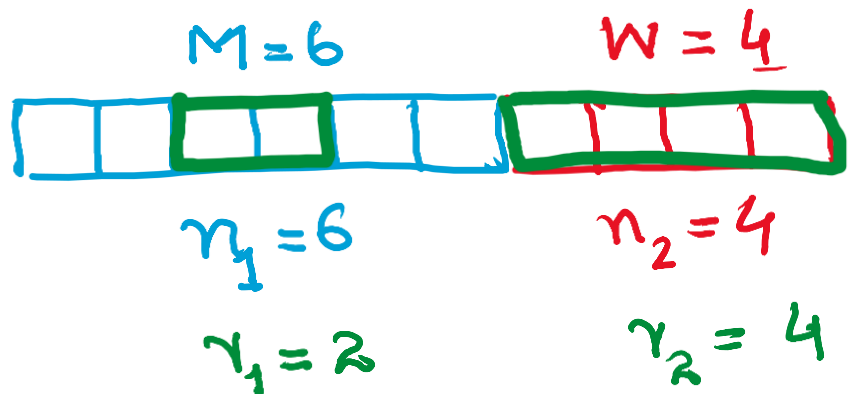


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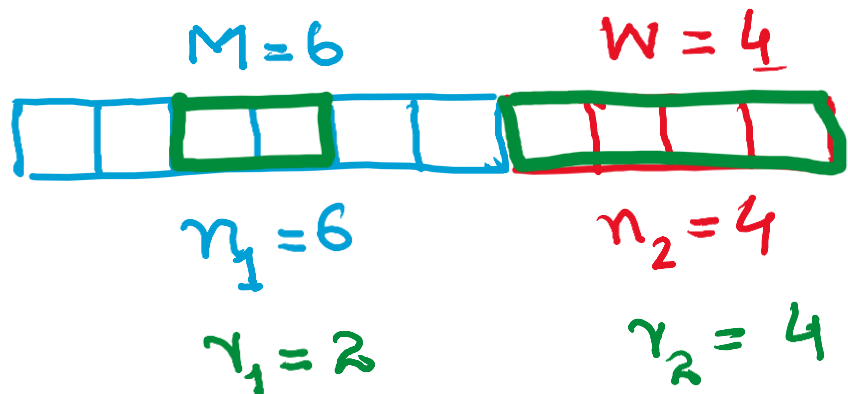


$$= {}^{n_1}C_{r_1} \times {}^{n_2}C_{r_2}$$



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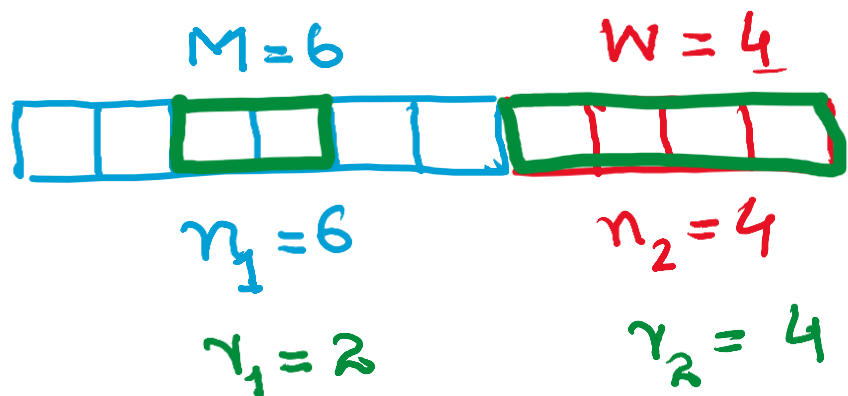
$$= {}_{n_1}C_{r_1} \times {}_{n_2}C_{r_2}$$

$$= {}_6C_2 \times {}_4C_4$$



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→ "constraint added"



$$= {}_{n_1}C_{r_1} \times {}_{n_2}C_{r_2}$$

$$= {}_6C_2 \times {}_4C_4$$

$$= {}_6C_2 \times 1$$

Ans



From **4 red**, **5 green**, and **6 yellow apples**, how many selections of **9 apples** are possible if 3 of each color are to be selected?

3 constraint



From **4 red**, **5 green**, and **6 yellow apples**, how many selections of **9 apples** are possible if 3 of each color are to be selected?

3 constraint

$$= {}^C_4 \times {}^C_5 \times {}^C_6$$



From **4 red**, **5 green**, and **6 yellow apples**, how many selections of **9 apples** are possible if 3 of each color are to be selected?

3 constraint

$$= {}^4C_3 \times {}^5C_3 \times {}^6C_3$$

Ans

Counting Sample points

✓ 1. Multiplication Rule

$$= n_1 \times n_2 \times \dots \times n_k$$

2. Combinations

✓ 2.1 "no constraints"

$${}_n C_r = \frac{n!}{r! (n-r)!}$$

Same

✓ 2.2 no constraints but "selection is made from different categories"

✓ 2.3 with constraints → "constraint selection from each different group"

$$= {}_{n_1} C_{r_1} \times {}_{n_2} C_{r_2} \times {}_{n_3} C_{r_3} \dots$$